Force Intellect CRM

Database Schema Proposal

Complete schema for all audited modules

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Based strictly on existing mock data in the application.
PostgreSQL-oriented types. Optional FKs are nullable to preserve current fuzzy / no-FK behavior.
Application source and mock data were NOT modified.
Source: src/data/*, src/auth/*, src/api/db.ts, src/api/validate.ts, src/api/policy.ts, src/api/session.ts

1. Lead Module Schema

Source: src/data/leads.ts (Lead, LEAD_REQUIREMENT_EXTRAS), src/api/validate.ts | Seed: 12 leads (LID-1001..1012)

No businessEntityId / contactId on Lead today. Matching uses normalizeOrgName fuzzy equality. business_entity_id and primary_contact_id below are proposed nullable columns.

Table: lead

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	VARCHAR(32)	Yes	Yes	-	e.g. LID-1001
2	lead_name	VARCHAR(255)	Yes	No	-	Often equals id in seed
3	company_name	VARCHAR(255)	Yes	No	-	Denormalized; required by API
4	business_entity_id	VARCHAR(32)	No	No	business_entity.id	Nullable proposal - no FK today
5	contact_person	VARCHAR(255)	Yes	No	-	Denormalized primary name
6	primary_contact_id	VARCHAR(32)	No	No	contact.id	Nullable proposal
7	email	VARCHAR(255)	Yes	No	-	Required by validateLeadWrite
8	phone	VARCHAR(64)	No	No	-	Optional if empty
9	lead_source	VARCHAR(64)	Yes	No	-	Typed LEAD_SOURCES (MasterData diverges)
10	lead_stage	VARCHAR(64)	Yes	No	-	LEAD_STAGES
11	lead_status	VARCHAR(64)	Yes	No	-	LEAD_STATUSES
12	products_required	TEXT	No	No	-	Comma-separated soft names
13	opportunity_value	NUMERIC(18,2)	Yes	No	-	Lead money field (not deal_value)
14	value_currency	CHAR(3)	No	No	-	Defaults INR; unused in seed
15	expected_closure_date	DATE	No	No	-	ISO date string today
16	assigned_salesperson	VARCHAR(255)	Yes	No	-	Name string, not user id
17	owner_user_id	VARCHAR(32)	No	No	app_user.id	Proposed until all owners are users
18	priority	VARCHAR(16)	No	No	-	High | Medium | Low; default Medium in UI
19	description	TEXT	No	No	-	Also leadDescription legacy field

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
20	last_activity_label	VARCHAR(64)	No	No	-	Relative text - prefer derived
21	created_at	DATE	Yes	No	-	Seed uses date-only
22	updated_at	TIMESTAMPTZ	No	No	-	Not found in app - optional

Table: lead_contact (from Lead.contacts[])

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	BIGSERIAL	Yes	Yes	-	LeadContactEntry has no id today
2	lead_id	VARCHAR(32)	Yes	No	lead.id	ON DELETE CASCADE
3	name	VARCHAR(255)	Yes	No	-	
4	email	VARCHAR(255)	No	No	-	
5	is_primary	BOOLEAN	Yes	No	-	primary flag
6	phones	TEXT[]	No	No	-	Or separate lead_contact_phone table

Table: lead_requirement_extra

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	lead_id	VARCHAR(32)	Yes	Yes	lead.id	1:1 from LEAD_REQUIREMENT_EXTRAS
2	approximate_quantity	TEXT	No	No	-	e.g. "3,200 user licenses"
3	requirement_details	TEXT	No	No	-	
4	remarks	TEXT	No	No	-	

Lead-scoped related tables (already in seed)

Table	FK to lead	Seed count	Source
activity	lead_id	12	src/data/activities.ts
meeting	lead_id	8	src/data/meetings.ts
document	related_id where related_type=lead	13	src/data/documents.ts

Table	FK to lead	Seed count	Source
note	related_lead_id	0 at seed	UI seedNotes on Lead Detail
email_message	lead_id	0 at seed	UI seedEmails on Lead Detail
opportunity	lead_id (optional)	4	src/data/opportunities.ts

Enums used by Lead

Field	Values
lead_stage	New; Contacted; Requirement Identified; Under Qualification; Qualified; Con...
lead_status	Open; On Hold; Cold; Converted; Dropped/Lost
lead_source (typed)	LinkedIn; Referral; Cold Call; Website; Trade Show; Email Campaign; Partner...
priority	High; Medium; Low

2. Opportunity Module Schema

Source: src/data/opportunities.ts, src/api/validate.ts | Seed: 4 opportunities (OPP-1001..1004) from converted leads

No opportunity-scoped children. Activities / meetings / notes / documents hang off lead_id. Documents relatedType is only lead | entity. weight_percent and weighted_value are derived from deal_stage.

Table: opportunity

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	VARCHAR(32)	Yes	Yes	-	e.g. OPP-1001
2	lead_id	VARCHAR(32)	No	No	lead.id	Optional for direct create
3	name	VARCHAR(512)	Yes	No	-	From displayLeadName on convert
4	company_name	VARCHAR(255)	Yes	No	-	No entity FK today
5	business_entity_id	VARCHAR(32)	No	No	business_entity.id	Nullable proposal
6	contact_person	VARCHAR(255)	Yes	No	-	Denormalized
7	email	VARCHAR(255)	Yes	No	-	
8	phone	VARCHAR(64)	No	No	-	
9	products_required	TEXT	No	No	-	
10	deal_value	NUMERIC(18,2)	Yes	No	-	From lead.opportunityValue on convert
11	value_currency	CHAR(3)	Yes	No	-	Seed always INR
12	deal_stage	VARCHAR(64)	Yes	No	-	DEAL_STAGES
13	status	VARCHAR(32)	Yes	No	-	Open | On Hold | Won | Lost
14	weight_percent	INTEGER	Yes	No	-	Derived from stage - prefer not store
15	weighted_value	NUMERIC(18,2)	Yes	No	-	Derived - prefer not store
16	closing_date	DATE	Yes	No	-	
17	owner	VARCHAR(255)	Yes	No	-	Name string
18	owner_user_id	VARCHAR(32)	No	No	app_user.id	Proposed
19	lead_source	VARCHAR(64)	No	No	-	Not in OPPORTUNITY_PATCH_KEYS

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
20	last_activity_label	VARCHAR(64)	No	No	-	Prefer derived
21	created_at	DATE	Yes	No	-	
22	created_by	VARCHAR(255)	Yes	No	-	
23	converted_at	TIMESTAMPTZ	No	No	-	Runtime convert only; absent on seed
24	converted_by	VARCHAR(255)	No	No	-	Absent on seed

Enums / stage weights

Field	Values
deal_stage	Qualification; Proposal; Negotiation; Commitment; Closed Won; Closed Lost
status	Open; On Hold; Won; Lost
deal_stage weight %	20 / 40 / 60 / 80 / 100 / 0

Seed records (4)

ID	Lead	Stage	Weight%	Deal value (INR)	Owner	Status
OPP-1001	LID-1007	Negotiation	60	1,20,00,000	Arjun Patel	Open
OPP-1002	LID-1006	Proposal	40	85,00,000	Vikram Nair	Open
OPP-1003	LID-1012	Qualification	20	2,50,00,000	Meena Iyer	Open
OPP-1004	LID-1005	Commitment	80	19,00,000	Priya Mehta	Open

3. Business Entity Module Schema

Source: src/data/businessEntities.ts, src/data/contacts.ts | Seed: 8 entities (BE-001..008), 12 catalog contacts (CON-001..012)

Dual contact models: nested EntityContact (C00x) is parallel to catalog Contact (CON-*). Lead has a third shape (LeadContactEntry). Recommend catalog contact as the CRM contact of record.

Table: business_entity

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	VARCHAR(32)	Yes	Yes	-	e.g. BE-001
2	name	VARCHAR(255)	Yes	No	-	Often includes Ltd. / Industries
3	type	VARCHAR(32)	Yes	No	-	EntityType
4	website	VARCHAR(255)	No	No	-	
5	gstin	VARCHAR(32)	No	No	-	
6	status	VARCHAR(32)	Yes	No	-	EntityStatus
7	products	TEXT	No	No	-	Free text
8	remarks	TEXT	No	No	-	
9	salesperson	VARCHAR(255)	No	No	-	Name from SALESPERSONS
10	owner_user_id	VARCHAR(32)	No	No	app_user.id	Proposed
11	created_date	DATE	Yes	No	-	
12	active_leads	INTEGER	No	No	-	Static seed - prefer derived
13	last_activity_label	VARCHAR(64)	No	No	-	Prefer derived

Table: entity_location

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	VARCHAR(32)	Yes	Yes	-	e.g. L001
2	business_entity_id	VARCHAR(32)	Yes	No	business_entity.id	
3	label	VARCHAR(128)	Yes	No	-	Head Office, Plant, Campus, etc.

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
4	address	TEXT	No	No	-	
5	city	VARCHAR(128)	No	No	-	
6	state	VARCHAR(128)	No	No	-	
7	country	VARCHAR(128)	No	No	-	
8	territory	VARCHAR(128)	No	No	-	From TERRITORIES list
9	preferred_port	VARCHAR(128)	No	No	-	
10	is_primary	BOOLEAN	Yes	No	-	

Table: entity_contact (nested on entity - parallel / legacy)

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	VARCHAR(32)	Yes	Yes	-	C001..C010 in seed
2	business_entity_id	VARCHAR(32)	Yes	No	business_entity.id	
3	name	VARCHAR(255)	Yes	No	-	
4	designation	VARCHAR(128)	No	No	-	
5	department	VARCHAR(128)	No	No	-	
6	mobile	VARCHAR(64)	No	No	-	
7	email	VARCHAR(255)	No	No	-	
8	is_primary	BOOLEAN	Yes	No	-	

Table: contact (catalog CONTACTS - preferred CRM contact)

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	VARCHAR(32)	Yes	Yes	-	CON-001..CON-012
2	business_entity_id	VARCHAR(32)	Yes	No	business_entity.id	Exists today
3	business_entity_name	VARCHAR(255)	No	No	-	Denormalized
4	name	VARCHAR(255)	Yes	No	-	
5	designation	VARCHAR(128)	No	No	-	

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
6	department	VARCHAR(128)	No	No	-	
7	mobile	VARCHAR(64)	No	No	-	
8	email	VARCHAR(255)	No	No	-	
9	location_label	VARCHAR(255)	No	No	-	Free text - not FK today
10	location_id	VARCHAR(32)	No	No	entity_location.id	Proposed
11	is_primary	BOOLEAN	Yes	No	-	
12	status	VARCHAR(16)	Yes	No	-	Active | Inactive
13	remarks	TEXT	No	No	-	
14	created_date	DATE	No	No	-	
15	last_activity_label	VARCHAR(64)	No	No	-	

Enums / lookups

Field	Values
type	Prospect; Customer; Dealer; Distributor; Other
status	Prospect; Customer; Inactive
contact.status	Active; Inactive
territory (11)	North India - Delhi NCR; Punjab & Haryana; West - Maharashtra; West - Gujar...

4. Auth Module Schema

Source: src/auth/users.ts, src/auth/types.ts, src/api/policy.ts, src/api/session.ts | Loginable users: 2

Credentials: password_hash EXISTS (SHA-256 hex). Plaintext demo passwords exist in DEV Login UI only - never store plaintext in DB. Values redacted in documentation as [REDACTED].

Table: app_user

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	VARCHAR(32)	Yes	Yes	-	USR-0042, USR-0001
2	name	VARCHAR(255)	Yes	No	-	
3	email	VARCHAR(255)	Yes	No	-	UNIQUE
4	role	VARCHAR(32)	Yes	No	-	Admin | Sales Manager
5	job_title	VARCHAR(128)	No	No	-	
6	initials	VARCHAR(8)	No	No	-	
7	password_hash	CHAR(64)	Yes	No	-	SHA-256 hex today; upgrade later
8	status	VARCHAR(32)	No	No	-	Not on AuthUser - optional Active
9	created_at	TIMESTAMPTZ	No	No	-	Not found on AuthUser

Seed loginable users (sensitive values redacted)

ID	Name	Email	Role	Job title	Initials	password_hash
USR-0042	Priya Mehta	priya.mehta@forceintellect.com	Sales Manager	Sales Manager	PM	[REDACTED]
USR-0001	Rajiv Anand	rajiv.anand@forceintellect.com	Admin	VP - Sales	RA	[REDACTED]

Table: auth_session (from StoredSession)

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	id	UUID	Yes	Yes	-	Proposed server session id
2	user_id	VARCHAR(32)	Yes	No	app_user.id	sub today

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
3	expires_at	TIMESTAMPTZ	Yes	No	-	TTL 8 hours today
4	nonce	VARCHAR(64)	Yes	No	-	32 hex chars
5	signature	VARCHAR(128)	No	No	-	Client stand-in today

Table: role_permission (from POLICIES)

Sl.No	Column Name	Data Type	Not Null	PK	Foreign Key	Remarks
1	role	VARCHAR(32)	Yes	Yes	-	Composite PK with permission
2	permission	VARCHAR(64)	Yes	Yes	-	crm.read, crm.write, ...
3	allowed	BOOLEAN	Yes	No	-	Or omit row if denied

Permission matrix (exact)

Permission	Who
crm.read	Any authenticated user
crm.write	Any authenticated user
audit.read	Admin only
settings.write	Admin only

Auth storage / session notes

Item	Value
sessionStorage key	fi-crm-session
StoredSession fields	sub, exp, nonce, sig
SESSION_SALT	fi-crm-session-v1 (hardcoded client stand-in)
Rate limit	In-memory Map; 5 attempts / 5 min per email
Remember-me	Not found
JWT / refresh token	Not found - session blob only

Item	Value
Roles (AuthRole)	Admin | Sales Manager
Salesperson roster vs auth	6 names in CRM; only 2 are loginable users